DATAWISE AI

DataWise AI: Project Presentation Report

Executive Summary, Product Overview, Key Differentiators & Live Demonstration Deck

Document Type:	Executive Presentation & Demonstration Deck
Target Audience:	Hackathon Evaluators, Executive Stakeholders, Technical Judges
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Repository Status:	Verified Codebase & Production Architecture
Orchestration:	LangGraph 16-Agent Autonomous Architecture

Architectural Foundation: Deterministic ML execution engine paired with autonomous multi-agent reasoning, strict data leakage boundaries, and human-in-the-loop validation.

1. Executive Summary & The Core Problem

DataWise AI is an Autonomous Data Science and Machine Learning Platform designed to eliminate the friction of modern predictive modeling. By coordinating an ensemble of 16 specialized AI agents, DataWise AI automates data cleaning, exploratory analysis, leakage prevention, algorithm selection, hyperparameter tuning, and production export.

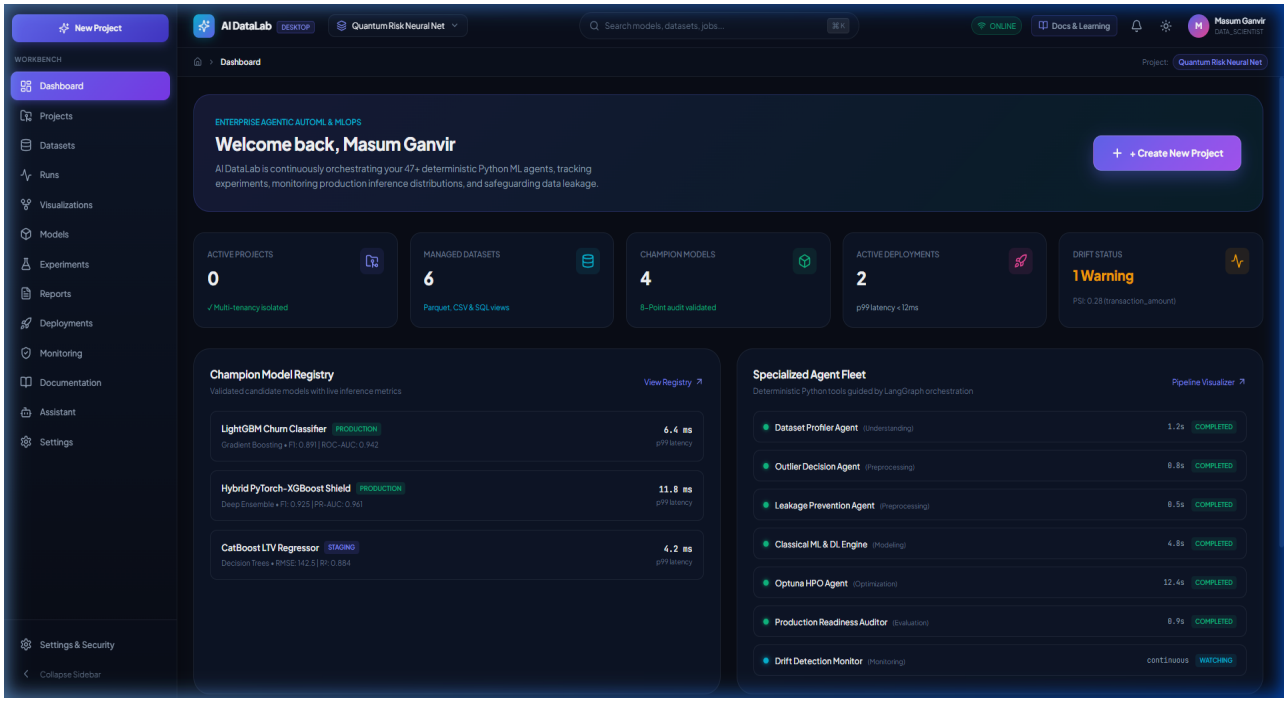


Figure: The DataWise AI Experience: Unified Executive Dashboard & Run Registry

2. Why DataWise AI Wins: Four Core Differentiators

1. **Deterministic Trust:** LLMs reason and explain; Python computes. No hallucinated metrics or imaginary models.
2. **Two-Prompt Simplicity:** Users describe their problem and requirements in plain English; the platform builds the technical roadmap.
3. **True Leakage Prevention:** Strict cross-validation discipline guarantees that validation scores reflect real production performance.
4. **Complete Artifact Delivery:** Users receive not just a model score, but serialized pipelines, Jupyter notebooks, and a deployable FastAPI server.

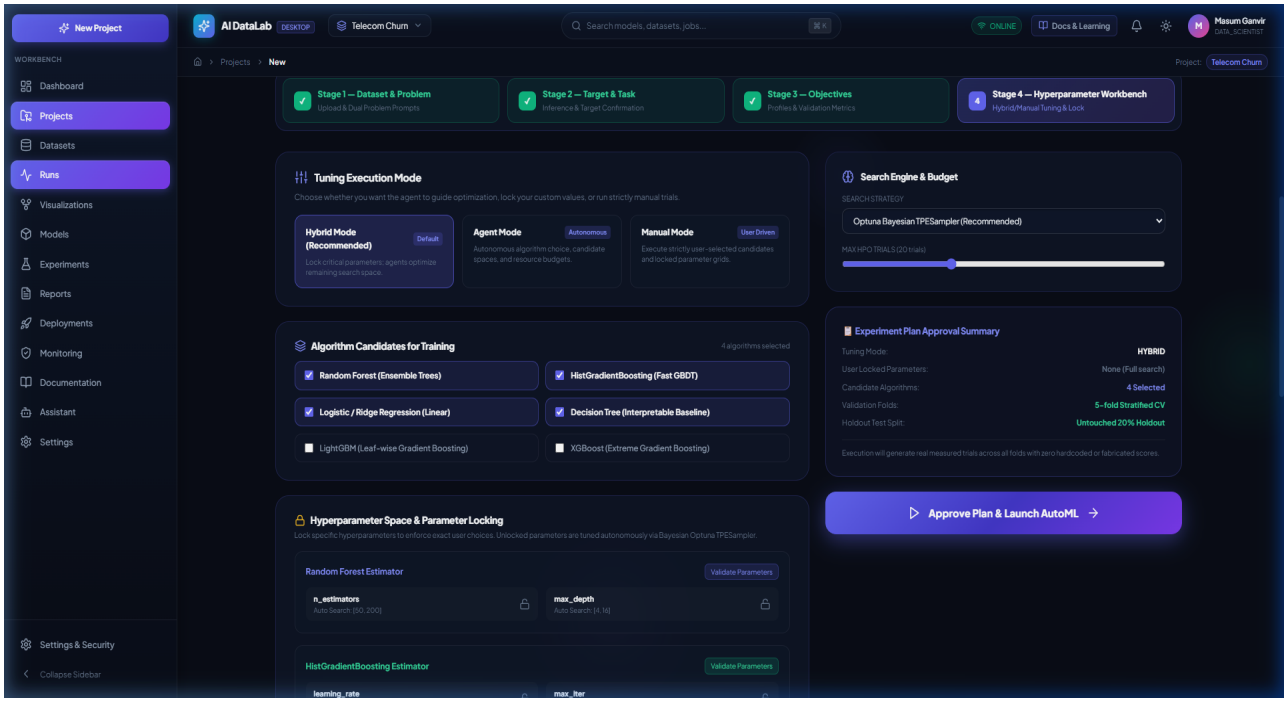


Figure: Interactive Live Demonstration: Stage 4 Agent Chat & Analysis Workbench

3. Live Demonstration Workflow (In 4 Steps)

Step	Platform Action	User Interaction & Outcome
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1. Upload & Scan	Instant schema validation and preview	User reviews summary metrics and missingness charts
2. Formulate Problem	Two-prompt guided problem capture	System confirms target column, metric, and constraints
3. Autonomous Run	16 agents profile, clean, engineer, & train	User monitors real-time execution stream and approves steps
4. Download Bundle	Compiles pipeline, notebook, report, and API	One-click download of production deployment package

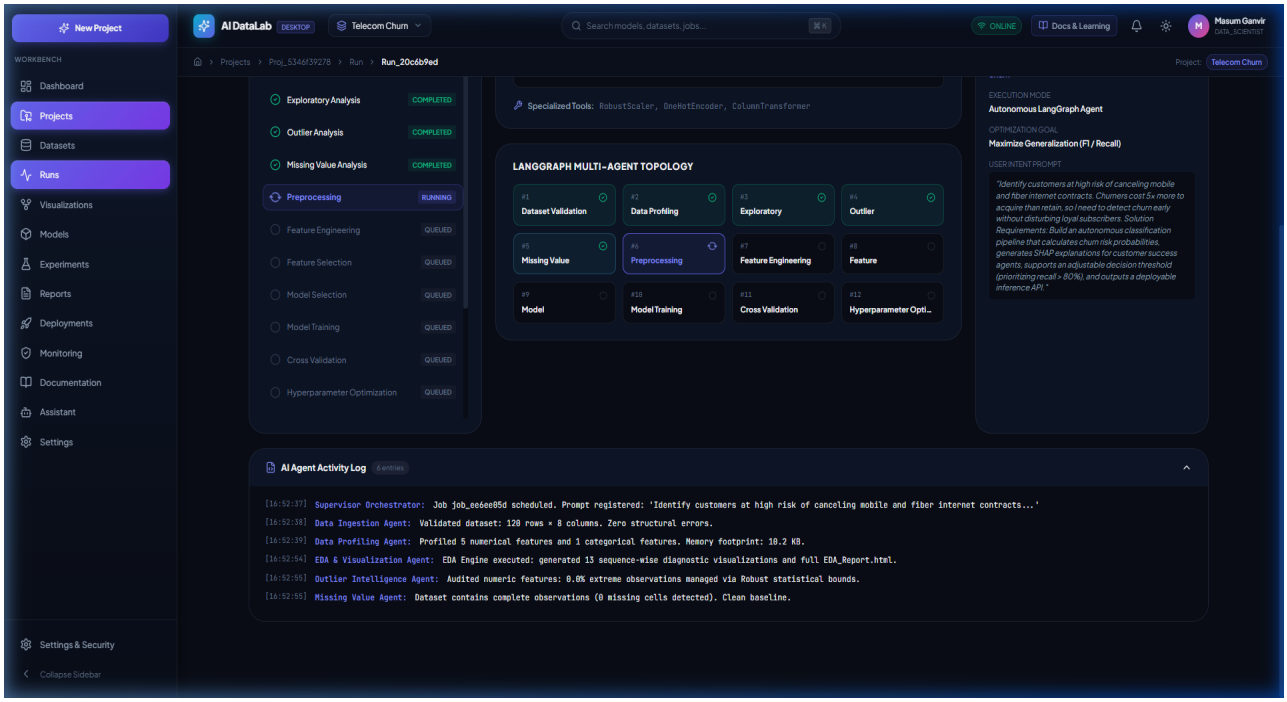


Figure: Live Autonomous Agent Execution Telemetry and Audit Trail